Gas Problem Set: Worked Solutions

Each solution follows the 5-step routine: **table → convert → pick equation → predict → solve/check**. Red = common mistake. R = 0.08206 L·atm/(mol·K), STP = 273.15 K and 1 atm (22.414 L/mol).

A. Pressure units & simple gas laws

1. 0.875 atm × 760 = **665 mm-Hg**; × 101.325 = **88.7 kPa**; × 1.01325 = **0.887 bar**. 642 mm-Hg ÷ 760 = **0.845 atm**.

2. `P₁=1.00 atm V₁=50.0 mL | P₂=? V₂=12.5 mL`; n and T constant → $P_1V_1 = P_2V_2$ → P_2 = 1.00 × 50.0/12.5 = **4.00 atm**. Prediction: volume ÷4 → pressure ×4 ✓. (mL is fine here because the units cancel.)

3. `T₁ = 295.15 K, T₂ = 77.15 K`. $V_2 = V_1(T_2/T_1)$ = 2.50 × 77.15/295.15 = **0.653 L**. Using °C gives a negative volume, which is impossible. Always use K!

4. $P_2 = P_1(T_2/T_1)$ = 32.0 × 318.15/288.15 = **35.3 psi**. Any pressure unit works for ratios; only T must be in K.

5. `P₁=755 mmHg V₁=5.00 L T₁=298.15 K | P₂=380. mmHg V₂=? T₂=258.15 K`, n constant.
$V_2 = V_1 \times (P_1/P_2) \times (T_2/T_1)$ = 5.00 × (755/380.) × (258.15/298.15) = 5.00 × 1.987 × 0.8658 = **8.60 L**.
Pressure dropped by about half (×1.99 for V), but T dropped only 13% (×0.866). The pressure effect wins, so the balloon expands. **Trick: write each change as a ratio factor and ask whether it should make V bigger or smaller.**

6. V/n constant: V_2 = 3.00 L × (0.175/0.125) = **4.20 L**. n_2 is the TOTAL (0.125 + 0.0500), not 0.0500.

B. PV = nRT

7. n = PV/RT = (150.)(15.0)/(0.08206 × 295.15) = **92.9 mol O_2** × 31.998 g/mol = 2973 g = **2.97 kg**.

8. n = 4.50/44.009 = 0.1023 mol; T = 310.15 K. P = nRT/V = (0.1023)(0.08206)(310.15)/2.00 = **1.30 atm**.

9. T = PV/(nR) = (2.00)(22.4)/(1.00 × 0.08206) = **546 K = 273 °C**. Check: at STP 1 mol in 22.4 L is at 1 atm, so doubling P at constant V means doubling T (273 K → 546 K) ✓.

C. Density & molar mass

10. $d = P\mathcal{M}/(RT)$: STP: (1.00)(70.90)/(0.08206 × 273.15) = **3.16 g/L**; 100 °C: (1.00)(70.90)/(0.08206 × 373.15) = **2.32 g/L**. Hotter gas → same mass spread over a larger volume → less dense ($d \propto 1/T$).

11. Empirical: C 85.6/12.011 = 7.127, H 14.4/1.008 = 14.29 → 1 : 2.00 → CH_2 (14.027 g/mol).
`m=0.802 g, V=0.354 L, T=373.15 K, P=752/760=0.9895 atm`
$\mathcal{M}$ = mRT/(PV) = (0.802)(0.08206)(373.15)/[(0.9895)(0.354)] = **70.1 g/mol**. k = 70.1/14.027 = 5.00 → **C_5H_{10}**.
Exam combo: % composition → empirical formula, gas data → $\mathcal{M}$, then k = $\mathcal{M}/\mathcal{M}$_emp.

12. At the same T and P, $d \propto \mathcal{M}$: **He (4) < air (29) < CO_2 (44)**. He balloons float; CO_2-filled balloons sink.

D. Dalton & over water

13. n(He) = 2.00/4.0026 = 0.4997 mol; n(O_2) = 8.00/31.998 = 0.2500 mol. RT/V = (0.08206 × 298.15)/5.00 = 4.893 atm/mol.
P(He) = 0.4997 × 4.893 = **2.45 atm**; P(O_2) = 0.2500 × 4.893 = **1.22 atm**; P_tot = **3.67 atm**. χ(He) = 0.4997/0.7497 = **0.667** (check: 0.667 × 3.67 = 2.45 ✓).
The partial pressure depends on MOLES, not grams: 8 g O_2 gives less pressure than 2 g He.

14. P(O_2) = χ·P_tot = 0.2095 × 0.950 = **0.199 atm = 151 mm-Hg**.

15. P(O_2) = 758 − 19.8 = 738.2 mm-Hg = 0.9713 atm; T = 295.15 K; V = 0.512 L.
n(O_2) = (0.9713)(0.512)/(0.08206 × 295.15) = 0.02053 mol → × (2 $KClO_3$/3 O_2) = 0.01369 mol → × 122.55 = **1.68 g $KClO_3$**.
Two traps: forgetting to subtract the water vapor, and forgetting the 2:3 ratio.

E. Gas stoichiometry

16. n(N_2) = PV/RT = (1.10)(65.0)/(0.08206 × 298.15) = 2.922 mol → × (2/3) = 1.948 mol NaN_3 × 65.01 = **127 g NaN_3**.
Pattern: gas → mol via PV=nRT, then mole ratio, then grams.

17. Same T and P → **liters behave like moles**. 6.00 L N_2 would need 18.0 L H_2, but only 15.0 L is available → **H_2 limiting**. NH_3 = 15.0 × (2/3) = **10.0 L**. N_2 used = 15.0/3 = 5.00 L → **1.00 L N_2 left over**.

18. n(C_3H_8) = 10.0/44.097 = 0.2268 mol. (a) CO_2 = 3 × 0.2268 = 0.6803 mol × 22.414 L/mol = **15.2 L at STP**. (b) O_2 = 5 × 0.2268 = 1.134 mol; V = nRT/P = (1.134)(0.08206)(298.15)/1.00 = **27.7 L**.
22.414 L/mol is ONLY valid at STP (0 °C, 1 atm); at 25 °C use PV = nRT (24.47 L/mol).

19. n(Al) = 2.00/26.982 = 0.07412 mol (÷2 = 0.0371). n(Cl_2) = (1.00)(1.50)/(0.08206 × 298.15) = 0.06131 mol (÷3 = 0.0204) → **Cl_2 is limiting**.
$AlCl_3$ = 0.06131 × (2/3) = 0.04087 mol × 133.33 = **5.45 g $AlCl_3$**.

F. Graham & KMT

20. (a) rate(H_2)/rate(O_2) = √(31.998/2.016) = **3.98× faster**.
(b) rate(X)/rate(He) = √($\mathcal{M}$_He/$\mathcal{M}$_X) → 0.354 = √(4.0026/$\mathcal{M}$) → $\mathcal{M}$ = 4.0026/0.354² = **31.9 g/mol → O_2**.
(c) Time is the inverse of rate: t_X/t_N_2 = √($\mathcal{M}$_X/$\mathcal{M}$_N_2) → $\mathcal{M}$ = 28.014 × (191/105)² = **92.7 g/mol**.
Heavier gas = slower rate = LONGER time. Square both sides to solve for $\mathcal{M}$.

21. rate(NH_3)/rate(HCl) = √(36.461/17.031) = 1.463. In the same time, the distances are in the same ratio: d(NH_3) = 1.463/(1 + 1.463) × 100 cm = **59.4 cm from the NH_3 end** (the ring forms closer to the HCl end because HCl is heavier and slower).

22. u_rms = √(3RT/$\mathcal{M}$), R = 8.314 J/(mol·K), T = 298.15 K.
He: √(3 × 8.314 × 298.15/0.0040026) = **1.36 × 10³ m/s**; Xe: √(3 × 8.314 × 298.15/0.13129) = **238 m/s**.
(b) Equal speeds mean equal T/$\mathcal{M}$: T(O_2) = 298.15 × (31.998/4.0026) = **2.38 × 10³ K**.
$\mathcal{M}$ in kg/mol (4.0026 g → 0.0040026 kg). With g/mol you'd get an answer √1000 ≈ 31.6× too small.

23. KE_avg = 3/2 RT = 1.5 × 8.314 × 298.15 = **3.72 kJ/mol**; per molecule 3/2 k_B T = **6.17 × 10⁻²¹ J**. **Doesn't depend on the gas**: KE depends only on T. Heavier gases have the same KE but move slower.

24. (a) T ×2, V fixed: **P ×2, KE ×2, u_rms ×√2 (1.41)**. Faster molecules hit the walls harder and more often.
(b) V ÷2, T fixed: **P ×2, KE and u_rms unchanged**. Same speeds, but more molecules per wall area, so there are more collisions per second.

G. Barometers & real gases

25. (a) Height ∝ 1/ρ: 760 mm × 13.56/1.00 = 10 306 mm = **10.3 m**. (b) P = ρgh = (1000 kg/m³)(9.80665 m/s²)(10.0 m) = 9.81 × 10⁴ Pa ÷ 101 325 = **0.968 atm**.

26. (a) **NH_3**: it is polar and H-bonds (strong attractions) and is a larger molecule; He is the most ideal gas. (b) **300 atm, 150 K**. The failed assumptions are **(1) particles have no volume** (fails at high P) and **(2) no interparticle forces** (fails at low T). Ideal gas: **Z = 1**.

H. Challenge

27. RT/V = (0.08206 × 400.)/2.00 = 16.41 atm/mol.
Before: 0.350 mol → **P = 5.74 atm**.
Reaction (mol): CO limits (0.200 CO needs only 0.100 O_2). After: CO = 0, O_2 = 0.150 − 0.100 = 0.050 mol, CO_2 = 0.200 mol.
P(O_2) = 0.050 × 16.41 = **0.82 atm**; P(CO) = **0**; P(CO_2) = 0.200 × 16.41 = **3.28 atm**; P_tot = **4.10 atm**.
The pressure dropped because 3 mol of gas become 2 mol of gas (fewer particles in the same V and T). **Method: do the limiting-reactant bookkeeping in moles first, then convert each gas to a pressure.**